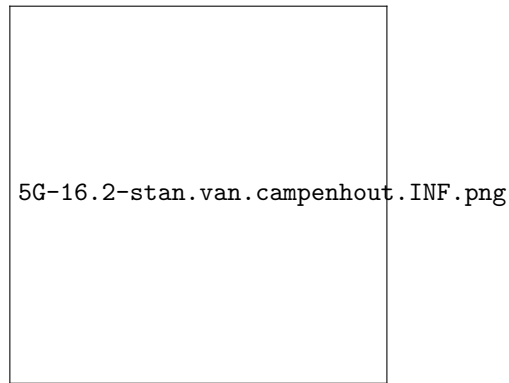


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$y = x^3$ rond $y = 1 \Rightarrow y = x^3 - 1$ rond de x-as over $[1, 3]$

$$\begin{aligned} \Rightarrow \pi \int_1^3 (x^3 - 1)^2 dx &= \pi \int_1^3 (x^6 - 2x^3 + 1) dx = \pi \left[\frac{x^3}{7} - \frac{x^4}{2} + x \right]_1^3 \\ &= \pi \left(\frac{3849}{14} - \frac{9}{14} \right) = \frac{1920\pi}{7} \end{aligned}$$

References